

FREDRIK PETTIT

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Education

University of British Columbia

Expected Graduation: May 2028

B.Sc. Computer Science (Artificial Intelligence) | GPA: 3.8

Vancouver, BC

Work Experience

AI Engineer Intern

Apr 2026 – Aug 2026

AEM

Langford, BC

- Shipped an AI IT support agent that automatically resolves **42%** of tickets across 9 brands, cutting an **800+** backlog
- Engineered the retrieval pipeline across bilingual docs, an LLM query-rewriting layer feeding hybrid vector and keyword search on Azure AI Search over OpenAI embeddings, returning grounded answers with citations and a **~2-3s** first token
- Cut monthly budget categorization from **6+ hours to 5 minutes** with a LangGraph agent that fixes vendors and GL codes across **1000+** transactions via Claude API, validated by an evaluation harness against finance's hand-done records

Software Engineering Intern

Jan 2026 – Apr 2026

ELLUSTROS

The Blue Mountains, ON

- Developed a recommendation engine that ranks **100+** office chairs against each user's measurements through layered filtering and weighted scoring across **15** ergonomic attributes, deployed as a FastAPI service powering a React Native app
- Cut validation defects per sprint by **32%** across the TRÖS Bespoke mobile app by writing **40+** Jest and React Testing Library tests covering user sign-in flows and AI-driven measurement capture, run in a CI/CD pipeline on each merge

Teaching Assistant | CPSC 310, CPSC 110

Aug 2025 – Present

UBC Department of Computer Science

Vancouver, BC

- Supported **900+** students in UBC's largest computer science course, achieving a 4.9/5 average student evaluation score
- Built 250+ randomized HTML/JavaScript questions and Python autograders for CPSC 310 - Intro to Software Engineering

Projects

YouTube Shorts AI Optimizer | *Python, FastAPI, AWS (Lambda, DynamoDB), OpenCV, XGBoost*

- Developed an end-to-end video optimization pipeline automating thumbnail selection, title, and description generation for YouTube Shorts, supporting a personal channel to grow to **38,000+** subscribers, **45,000,000+** views, and **1,800,000+** likes
- Trained four XGBoost regressors on **3000+** videos to score thumbnail, title, description, and hashtag candidates, predicting engagement rate at **0.043 MAE** from OpenCV/Rekognition thumbnail features and sentence-transformer text embeddings
- Orchestrated the pipeline as an asynchronous AWS Step Functions workflow, fanning video-analysis and inference jobs across Lambda functions via SQS queues, then calling Gemini 1.5 Pro for frame selection and Claude API for text generation

GitIntent - UC Berkeley AI LLM Hackathon 2026 | *Python, TypeScript, FastAPI, Redis Stack (Redisearch), voyage-code-3*

- Built a system that lets AI coding agents learn from developers and fellow agents by recording every change with the prompt behind it, then recalling failed attempts before repeating them, lifting performance **8.3%** across **36** DeepSWE runs
- Designed the hybrid retrieval engine on Redis Stack, fusing 1024-dimensional voyage-code-3 vector search (HNSW) with BM25 keyword ranking through Reciprocal Rank Fusion, deduplicating matches up to snapshots to reach **~93%** recall@5
- Engineered a content-addressed hashing scheme identical on client and server, giving idempotent, deduplicated writes

Inventory & Sales Automation Platform | *Spring Boot, React, TypeScript, PostgreSQL, Python, FastAPI*

- Built and operate custom infrastructure for a personal cross-platform retail business on Grailed, Depop, and eBay, logging 60+ international orders and **\$6K+** in transactions through Spring Boot API, React/TypeScript frontend, and PostgreSQL
- Automated order logging by parsing Gmail sale confirmations into the tracker, and built a Python/FastAPI Playwright service that reposts 15 listings weekly on a 30-day bump-then-reduce price cycle to maximize buyer visibility
- Improved query performance 9× across 10+ aggregated reporting views by adding targeted indexes, powering the dashboards of a polyglot backend that pairs a Spring Boot cashflow API with a Python/FastAPI Playwright sidecar over JSON/HTTP

Technical Skills

Languages & Databases: Java, Python, C++, C#, SQL, R, JavaScript/TypeScript, PostgreSQL, DynamoDB, Redis

AI/ML: PyTorch, TensorFlow, scikit-learn, XGBoost, OpenCV, sentence-transformers, embeddings, hybrid retrieval, RAG

LLM & Agents: LangGraph, LangChain, LangSmith, OpenAI, Claude, Gemini APIs, voyage-code-3, evals

Frameworks: Spring Boot, FastAPI, React, React Native, Expo, JUnit, Jest, Playwright, Pydantic, pandas, NumPy

Cloud & DevOps: AWS (Lambda, Step Functions, SQS, Rekognition), Azure AI Search, Docker, Gradle, Git, CI/CD